

Cahit Ikbal Er

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Professional Summary

Ph.D. candidate in Mechanical Engineering working on planning and control for heterogeneous multi-robot teams. I design scalable algorithms with provable guarantees on energy-awareness, robustness, feasibility, and complexity for long-horizon UAV-UGV missions such as aerial monitoring, coverage, and collaborative exploration, and validate them on real hardware.

Education

Northeastern University

PH.D. IN MECHANICAL ENGINEERING

Concentration: Mechatronics

Boston, MA

Expected: April 2028

Northeastern University

M.Sc. IN ELECTRICAL AND COMPUTER ENGINEERING

Concentration: Robotics

Boston, MA

April 2026

Bilkent University

B.Sc. IN MECHANICAL ENGINEERING

Ankara, Turkey

June 2023

Publications

- [1] **C. I. Er**, A. Kashiri, Y. Yazicioglu. **RSPECT: Robust and Scalable Planner for Energy-Aware Coordination of UAV-UGV Teams in Aerial Monitoring**. Journal of Intelligent & Robotic Systems (**JINT**), 2026. [paper] [video]
- [2] **C. I. Er**, S. Juttu, Y. Yazicioglu. **Energy-Aware Collaborative Exploration for a UAV-UGV Team**. International Conference on Intelligent Robots and Systems (**IROS 2026**). [preprint] [video]
- [3] R. Fowler*, **C. I. Er***, B. Johnsenberg, Y. Yazicioglu. **PRO-SPECT: Probabilistically Safe Scalable Planning for Energy-Aware Coordinated UAV-UGV Teams in Stochastic Environments**. International Conference on Intelligent Robots and Systems (**IROS 2026**). *Equal contribution. [preprint] [video]

Experience

Multi-Agent Robotics and Autonomy Lab (MuRAL), Northeastern University

Boston, MA

RESEARCH ASSISTANT

Sep. 2023 – present

- Build and maintain the group's UAV-UGV hardware testbed, including the two-team setup used in [1]: two Crazyflie UAVs and two TurtleBot4 UGVs under OptiTrack, integrated in ROS2.
- Integrated and deployed the exploration stack in [2] on a Crazyflie with multi-ranger deck and a TurtleBot4 with RPLIDAR, running occupancy mapping, frontier detection, and information-gain viewpoint selection in ROS2; the team reached 95% exploration of an unknown environment.
- Design and run ROS2 simulation environments for validating planning under uncertainty, including the stochastic wind-disturbance testbed used in [3].
- Formulate the underlying planning problems in [1], [3] as mixed-integer programs and design polynomial-time heuristics that scale to 100 monitoring points and 10 teams, with guarantees on feasibility, robustness, and worst-case complexity, and comparisons against Branch and Cut, simulated annealing, genetic algorithms, and published baselines.

Miniature Robotics Lab (MRL), Bilkent University

Ankara, Turkey

UNDERGRADUATE RESEARCHER

Feb. 2022 – July 2023

- Design, fabrication, and locomotion analysis of Miniature Soft Quadruped (SQQuad) robots.

AELSAN

Ankara, Turkey

CANDIDATE ENGINEER

Dec. 2022 – June 2023

- Designed and optimized Kalman filters for motor control subsystems in SATCOM terminal and ground center systems, as part of broader mechatronics-oriented R&D.

Skills

Programming

Python, MATLAB, Julia, C++

Robotics Hardware

Crazyflie, TurtleBot4, OptiTrack, E-puck2

Robotics Software

ROS2, ROS, Gazebo, RViz, Gurobi, Simulink, Docker, Git, Linux

Planning & Control

Motion Planning, State Estimation, Multi-Agent Coordination, Task Allocation

Optimization & Theory

Mixed-Integer Programming, Vehicle Routing & Scheduling, Combinatorial Optimization